

Spherepop: The Algebra of the topology of Reachable Computation

A Complete Theory including
Type Systems, Error Correction,
and **Garbage Collection.**

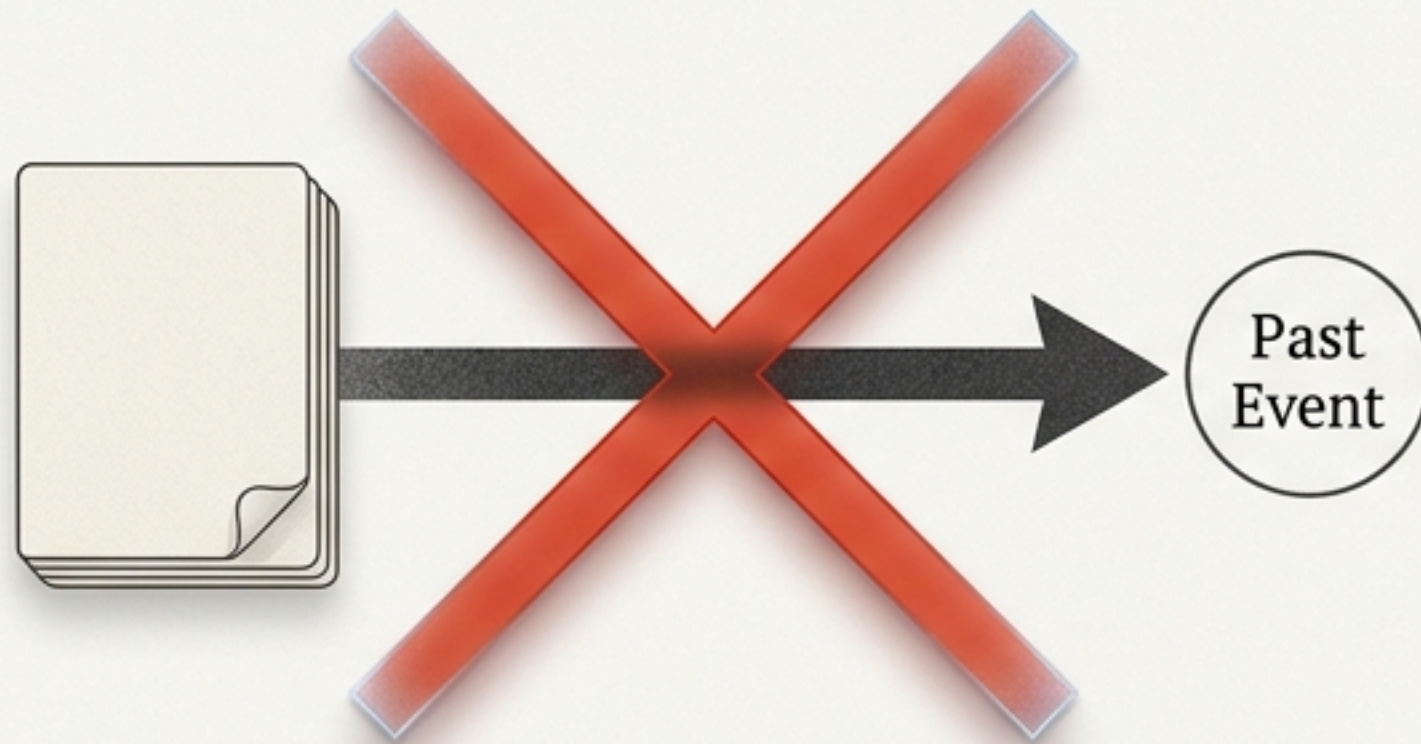
Flyxion, Independent Researcher.



THE TWO GREAT ERRORS OF REPRESENTATION

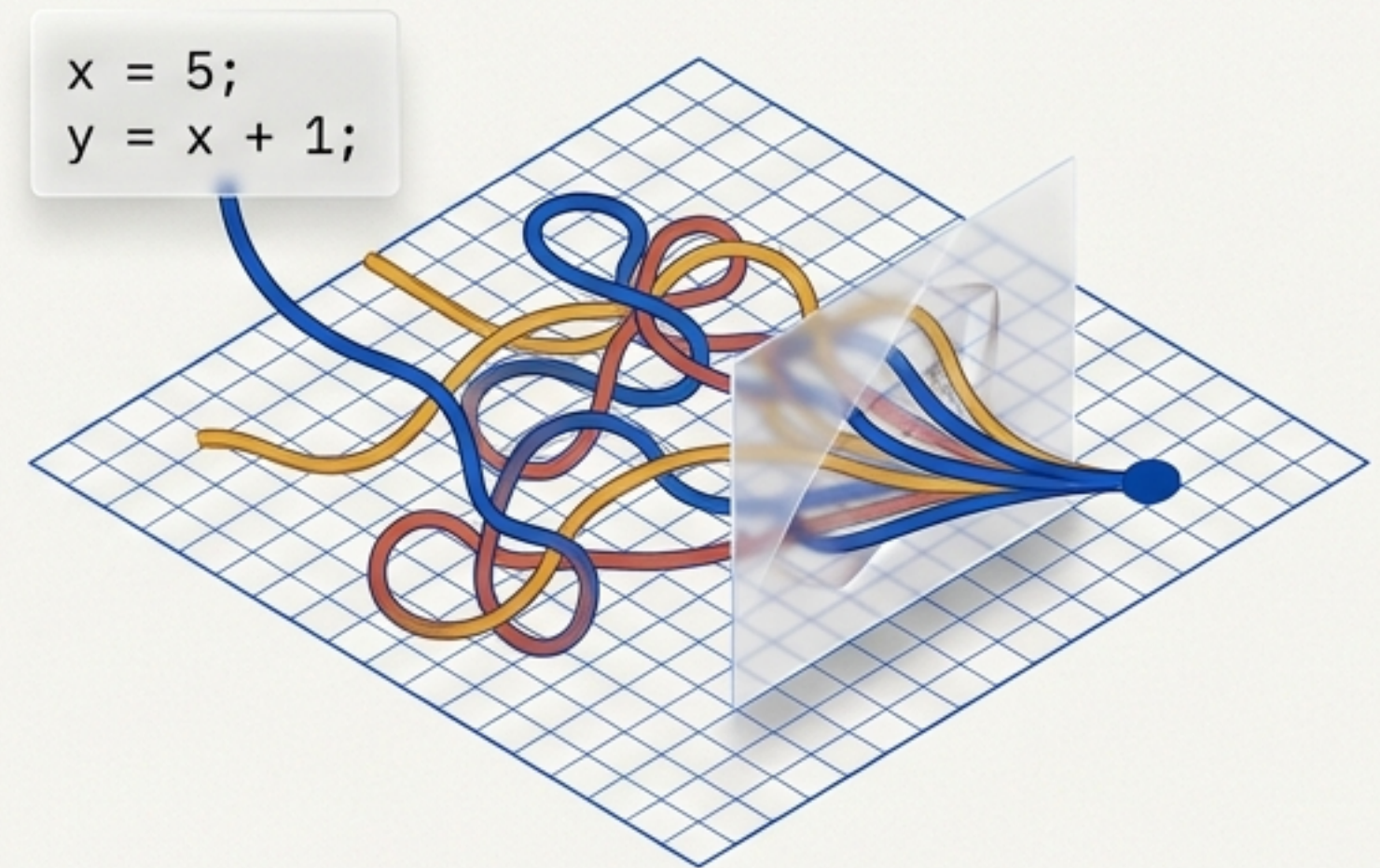
THE PASSIVE THEORY OF NOTES

Notes are **NOT** prosthetic memories recording past states.



THE STATE-CENTRIC ASSUMPTION

Programs do **NOT** merely mutate present values.



$$D(s) = |\sigma^{-1}(s)| > 1$$

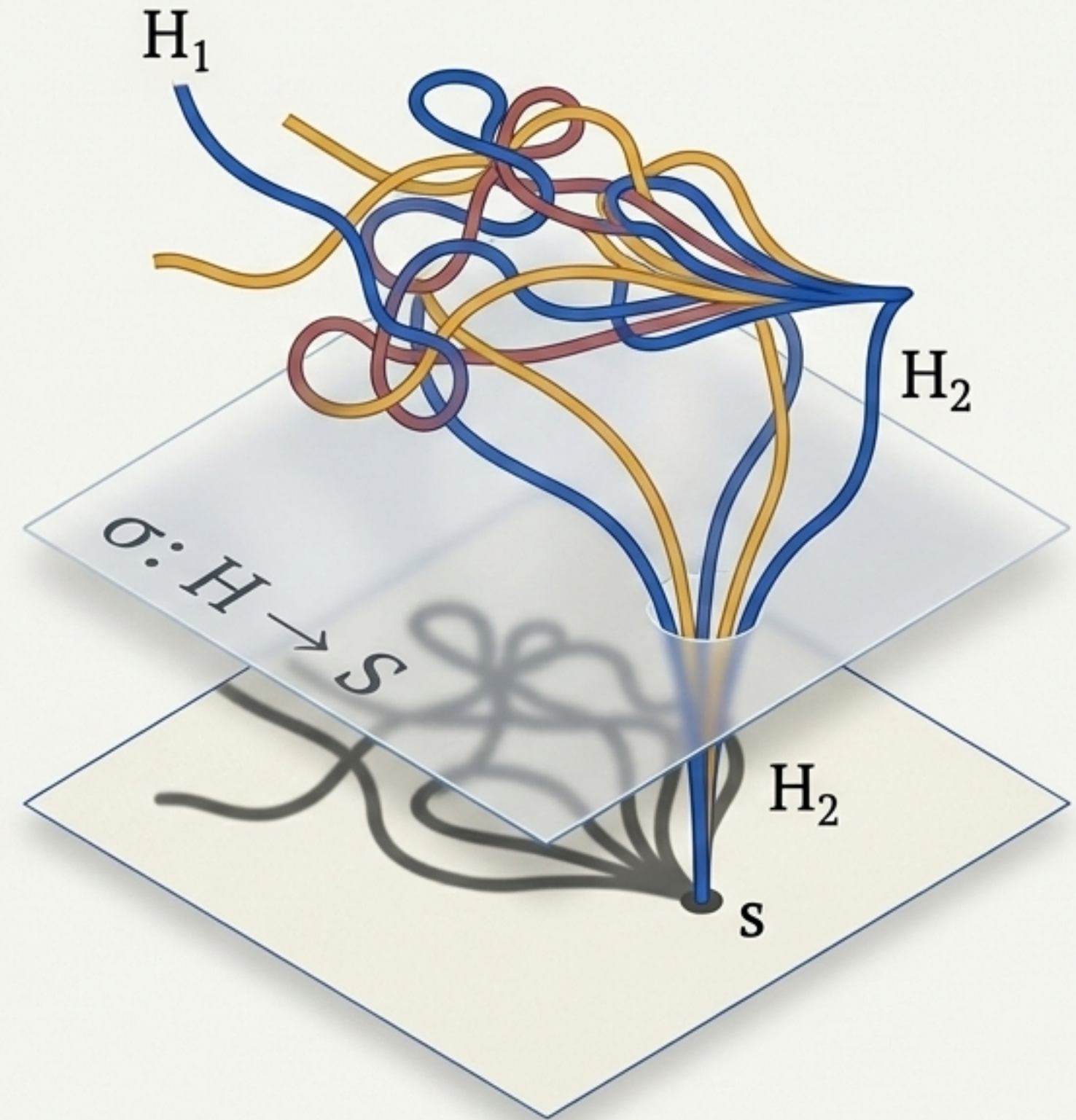
Both models suffer from State Degeneracy. They identify value with a terminal configuration rather than the historical trajectory that produced it, irreversibly discarding causal truth.

The Spherepop Inversion: History is Ontologically Prior to State

State is not primary. State is a quotient—a compressed residue.

The State Illusion occurs when we mistake the 2D projection for the reality of the multi-dimensional history. If history is what is lost in traditional computing, **history** must be what is **preserved**.

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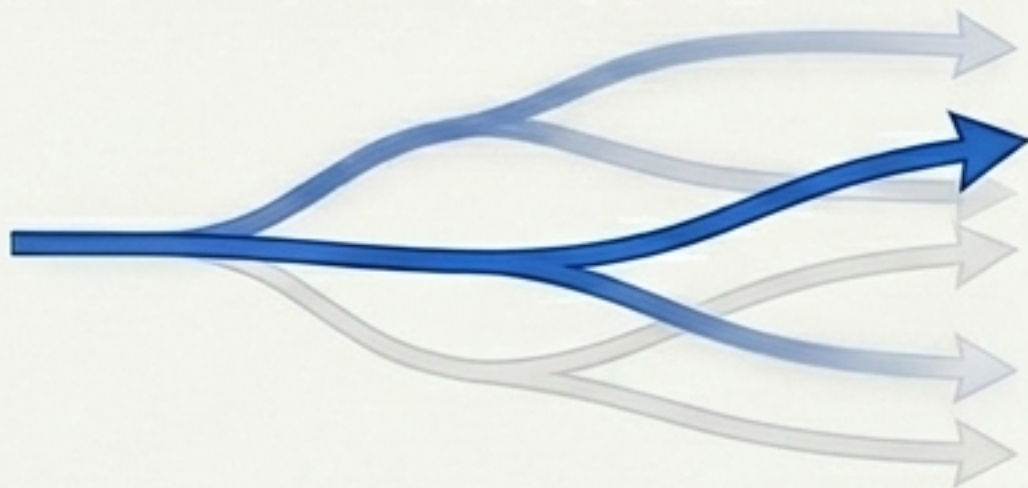


THE CALCULUS OF POSSIBILITY: FOUR IRREDUCIBLE OPERATORS

These four operators form the minimal complete set for managing possibility space. Variable assignment, conditionals, application, and deallocation are entirely derived from these primitives.

POP (COMMITMENT)

$$(H, \Omega) \rightarrow (H ++ [\text{Pop}(x)], \Omega \setminus \{x\})$$



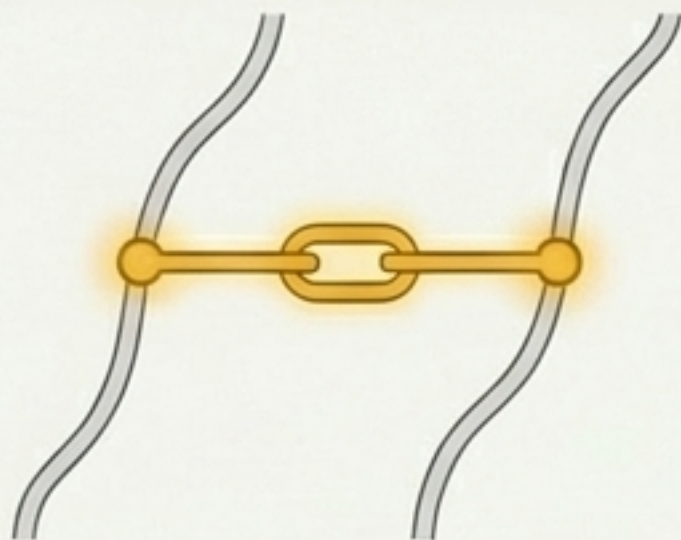
REFUSE (DOCUMENTED INADMISSIBILITY)

$$(H, \Omega) \rightarrow (H ++ [\text{Refuse}(x, r)], \Omega)$$



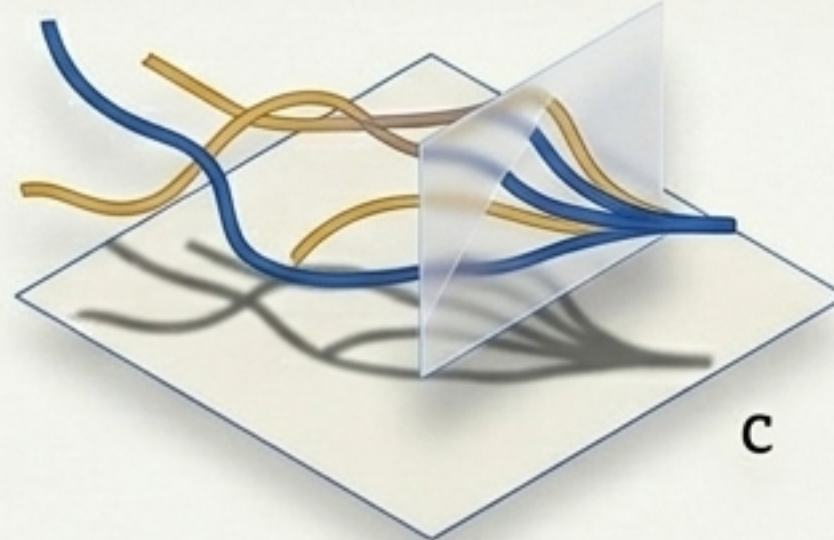
BIND (DEPENDENCY)

$$(H, \Omega) \rightarrow (H ++ [\text{Bind}(a, b)], \Omega)$$



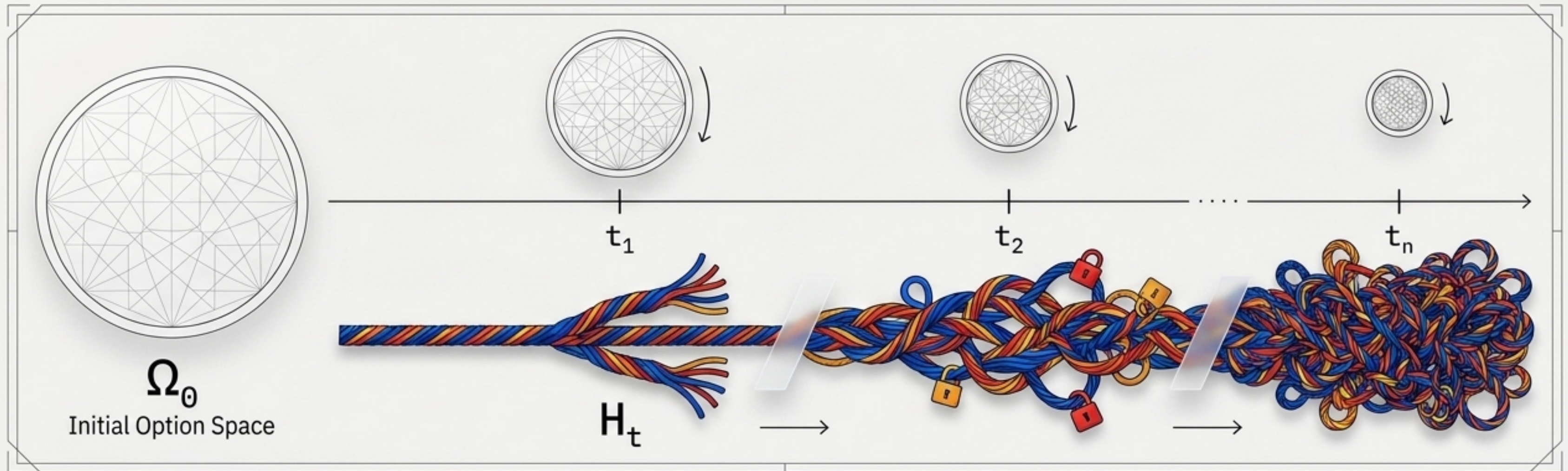
COLLAPSE (OBSERVATION)

$$(H, \Omega) \rightarrow (H ++ [\text{Collapse}(x, c)], \Omega)$$



The Conservation Law of Possibility

Possibility is never created or destroyed; it is only consumed by Pop or documented as inadmissible by **Refuse**. Because the event weight history strictly increases ($|H_{t+1}| > |H_t|$), execution is fundamentally irreversible. There is no return to a prior state without negating computation itself.

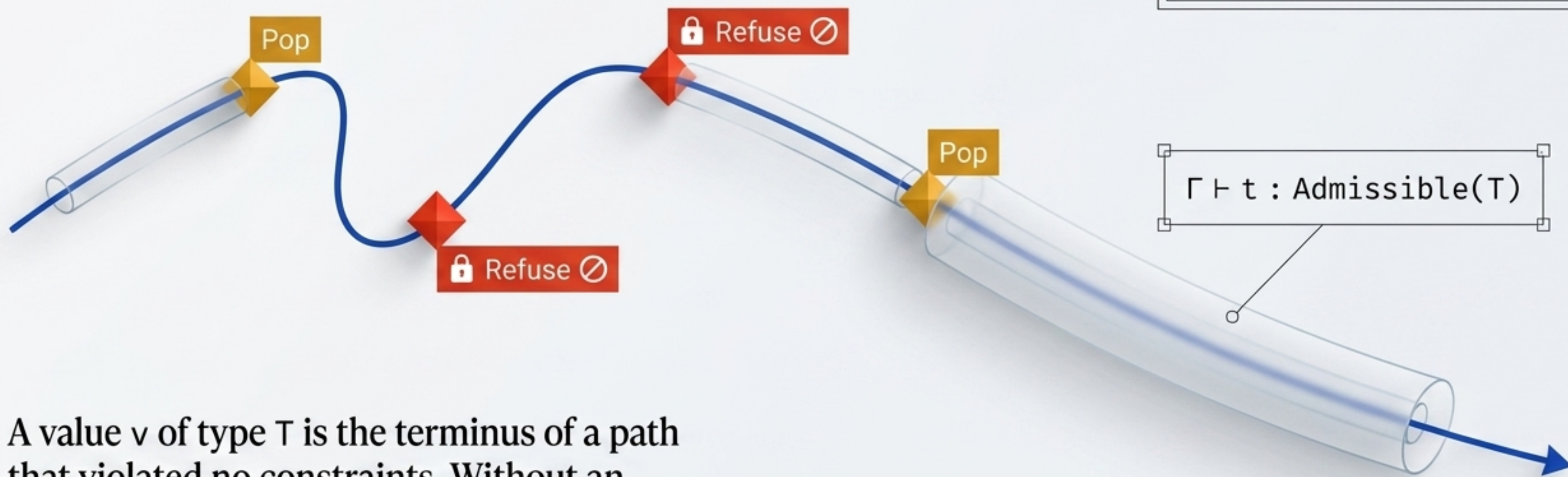


$$\Pi(H_t, \Omega_t) = |\Omega_0|$$

Types as Admissibility Certificates

VIEW CONTRAST

- The **Standard View**: A type is a set of valid values.
- The **Spherepop View**: A type is a proof of admissible history.



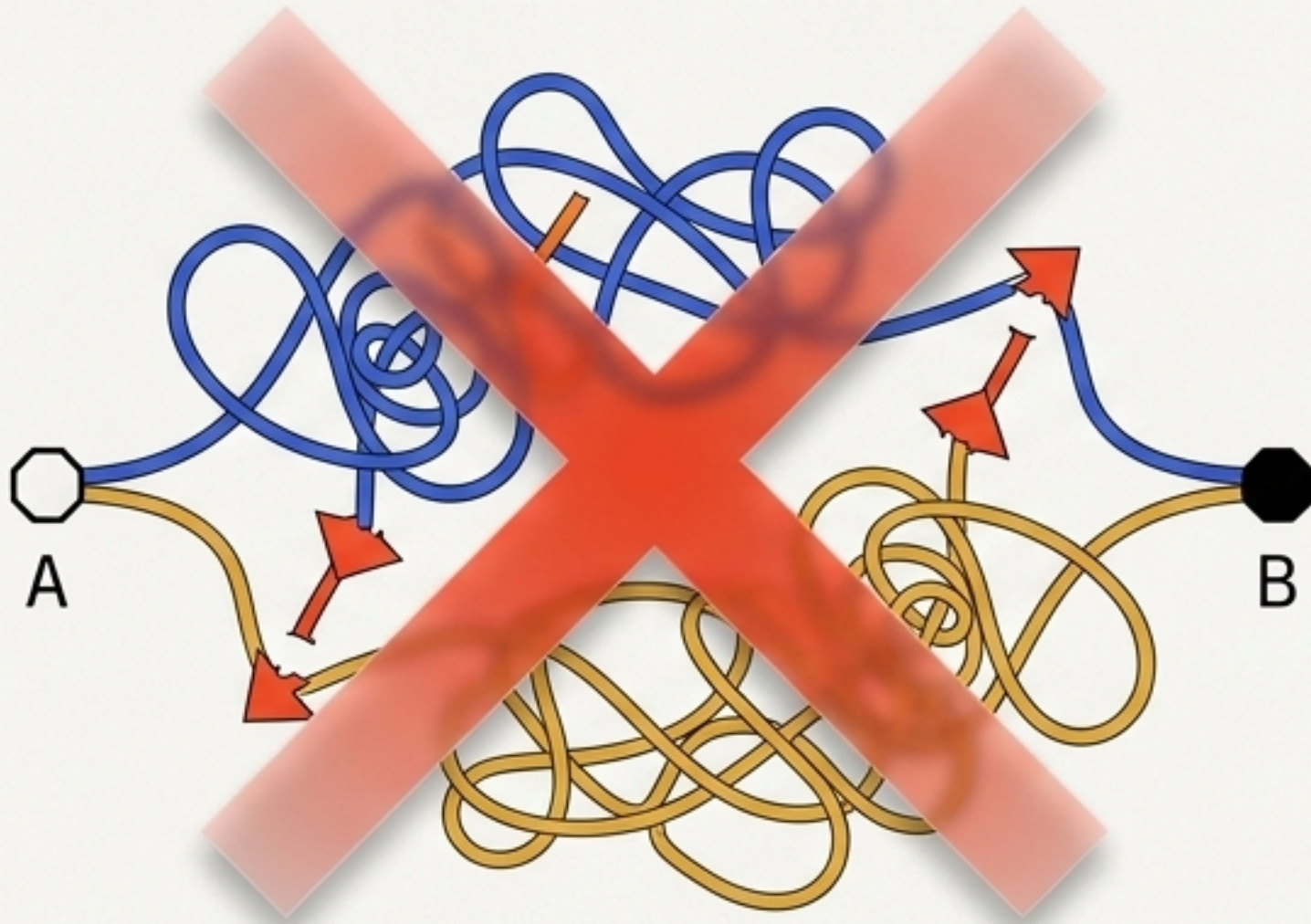
$\Gamma \vdash t : \text{Admissible}(T)$

A value v of type T is the terminus of a path that violated no constraints. Without an $\text{Admissible}(T)$ certificate, a Collapse operation triggers a hard type error, not a runtime exception.

Event IR and the Fallacy of Output Equivalence

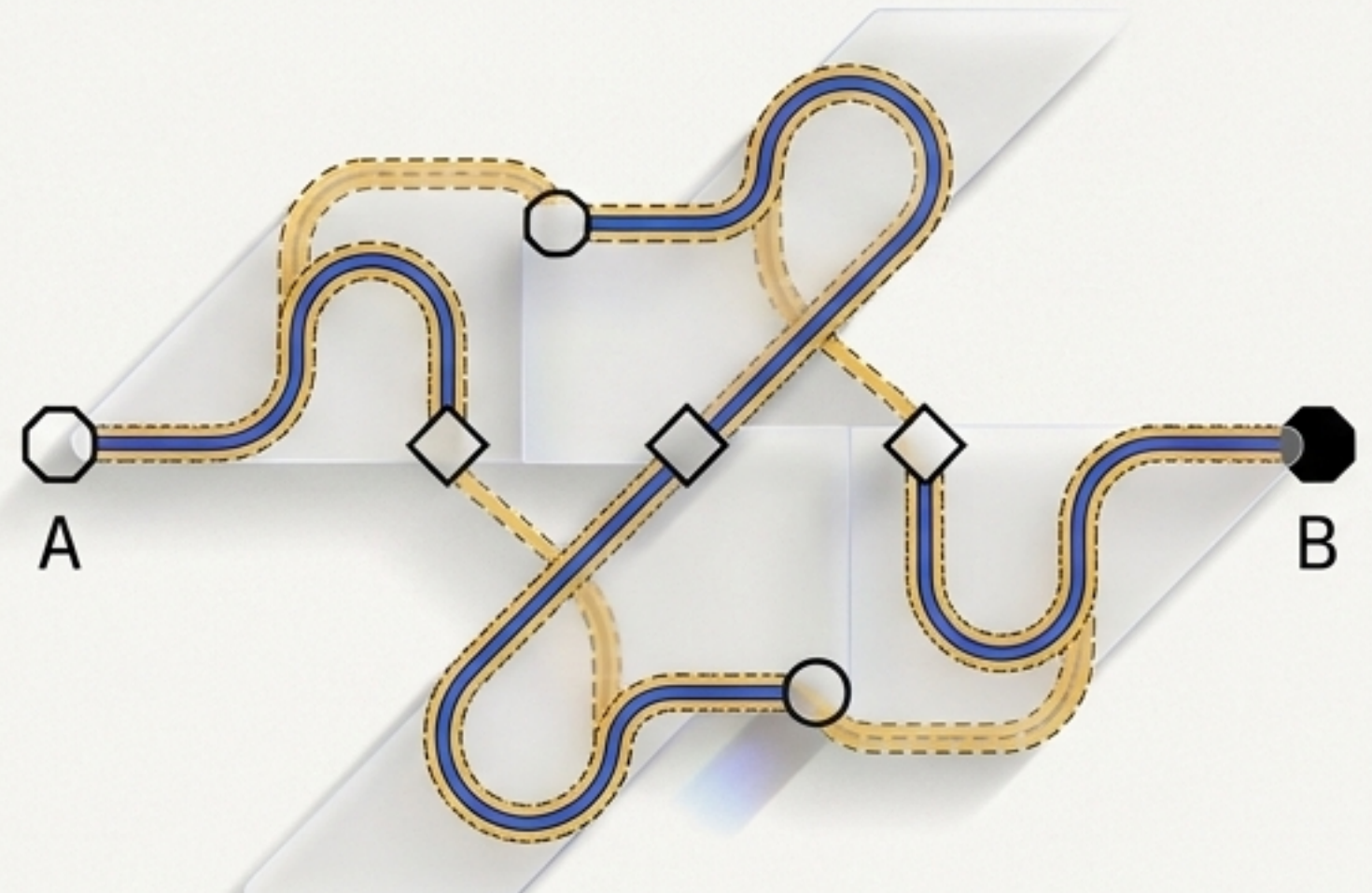
Traditional compilers mutate machine state until the final output matches. Spherepop's Event IR records events. A compiler is only correct if the sequence of primitive events generated by the compiled form factorizes uniquely and identically to the interpreter's history.

Output Equivalence - FAILED



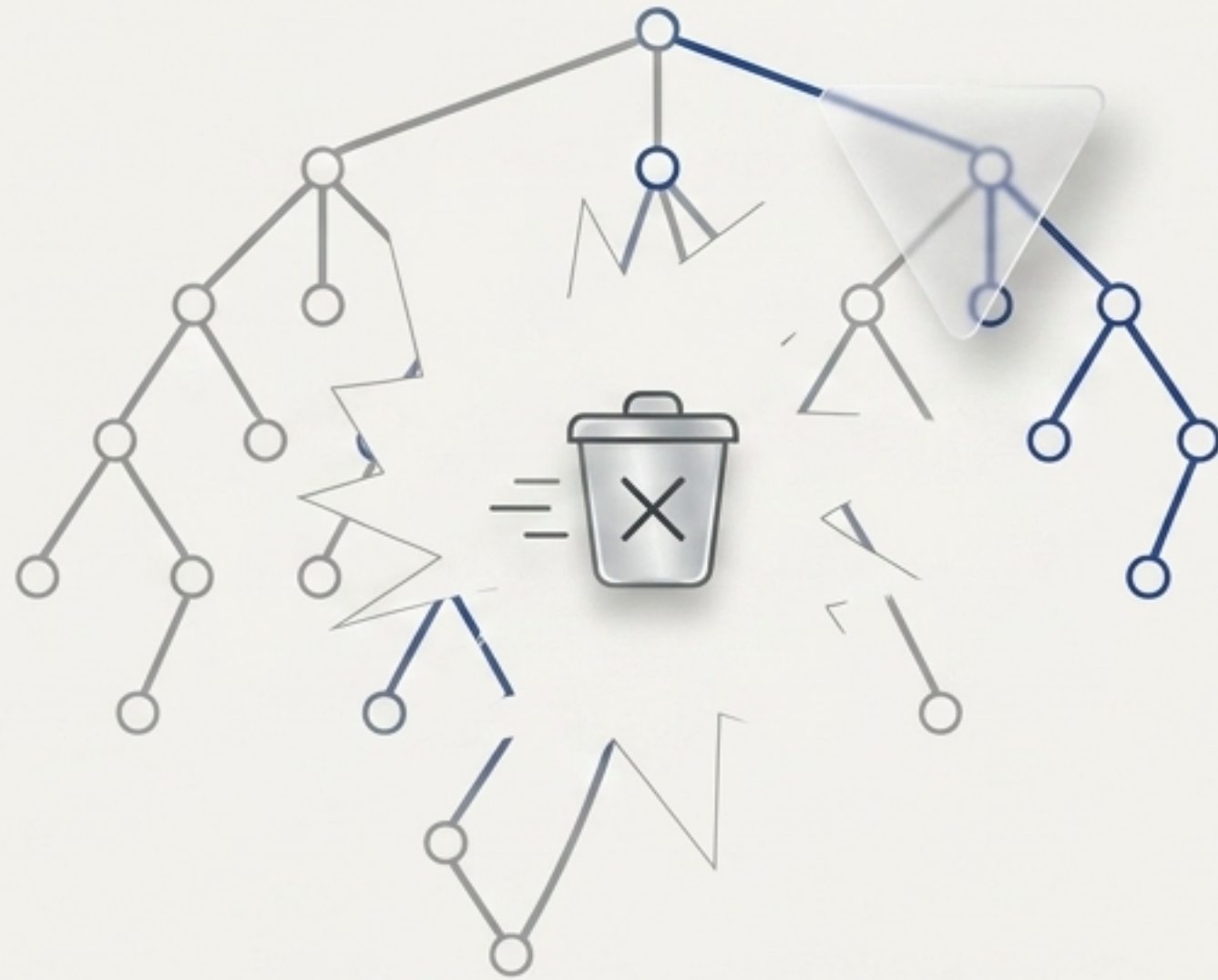
Replay Equivalence - VERIFIED ✓

$$I(p).history \equiv V(C(p)).history$$

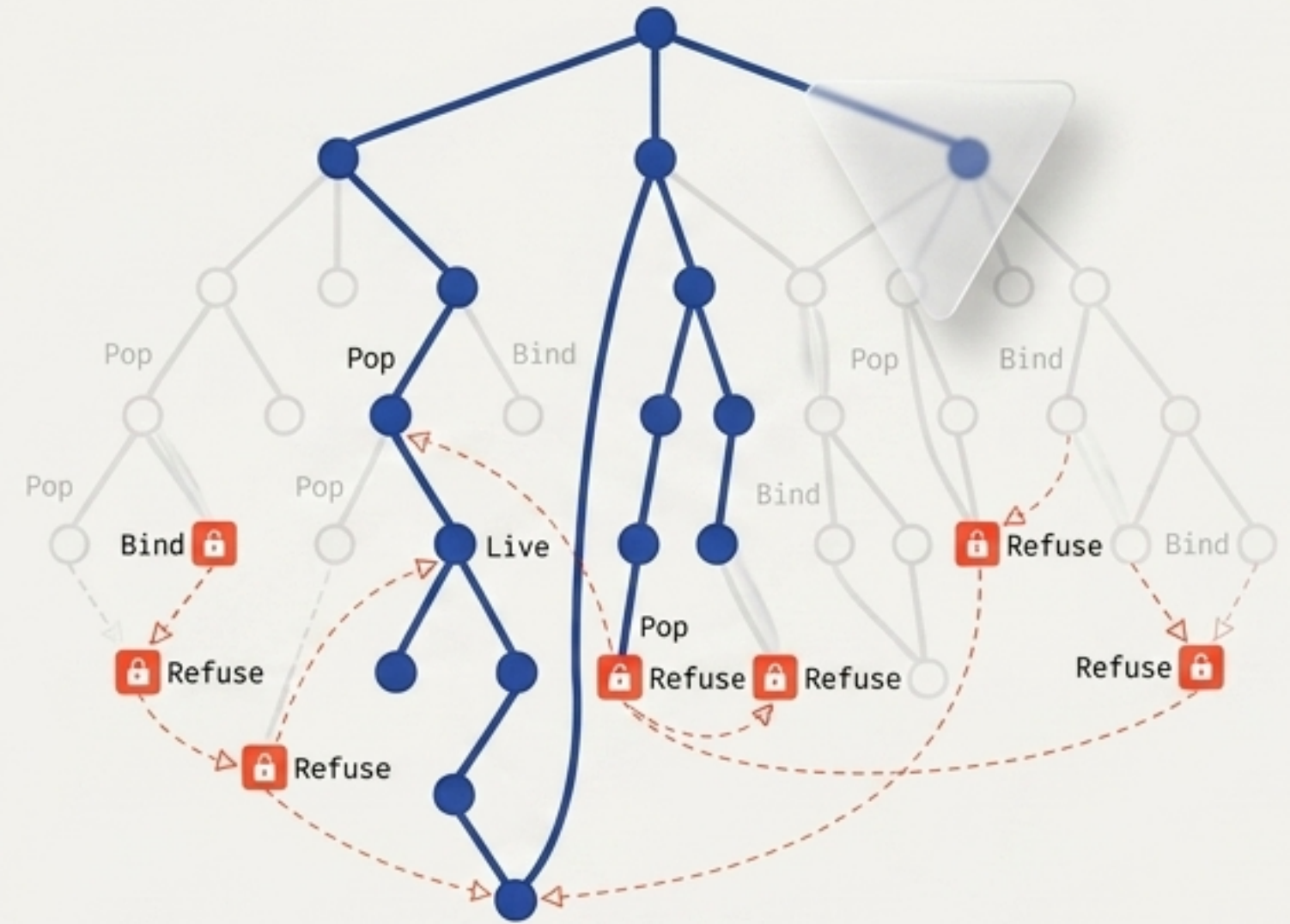


Admissibility-Guided Garbage Collection

Standard GC



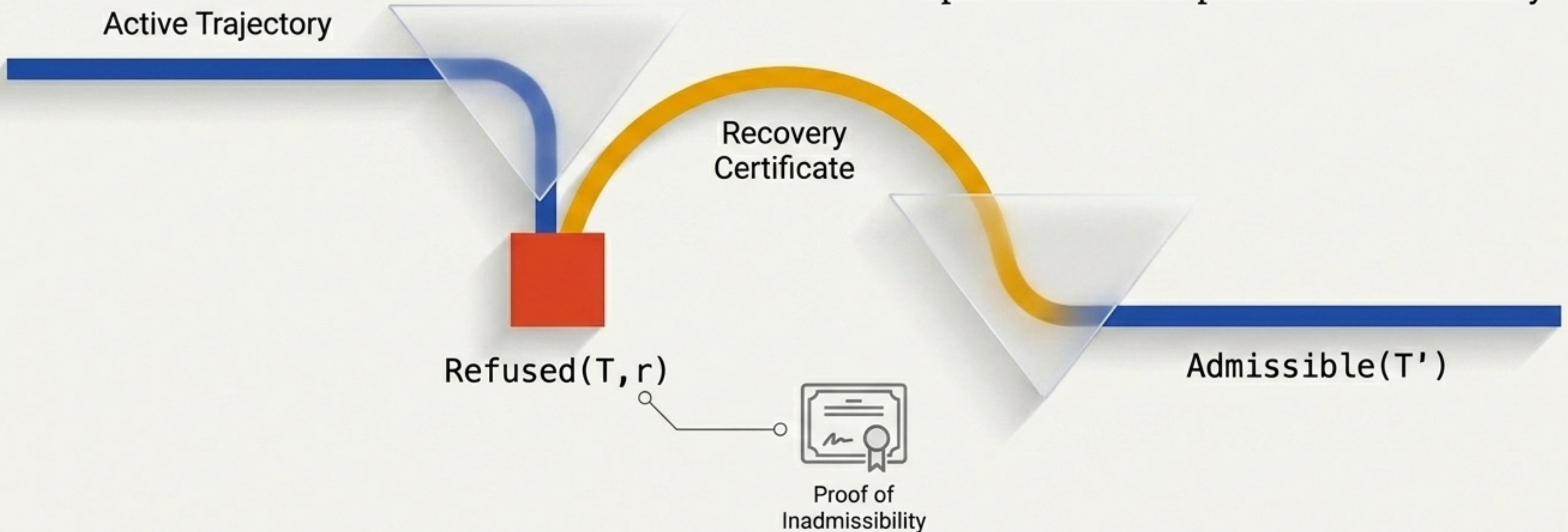
Certificate-Only GC (Spheredpop)



Garbage is not defined by structural unreachability, but by historical irrelevance. Certificate-Only GC prunes dead operations while permanently retaining the Refuse events required to mathematically support pending collapse judgements.

Error Correction as Certified Refusal

Errors are not exceptions; they are structurally inevitable non-traversals. Because the history is preserved, error recovery is not a blind unwinding of the stack, but a certified continuation built upon documented proof of inadmissibility.



The Paradigm Shift: State vs. History

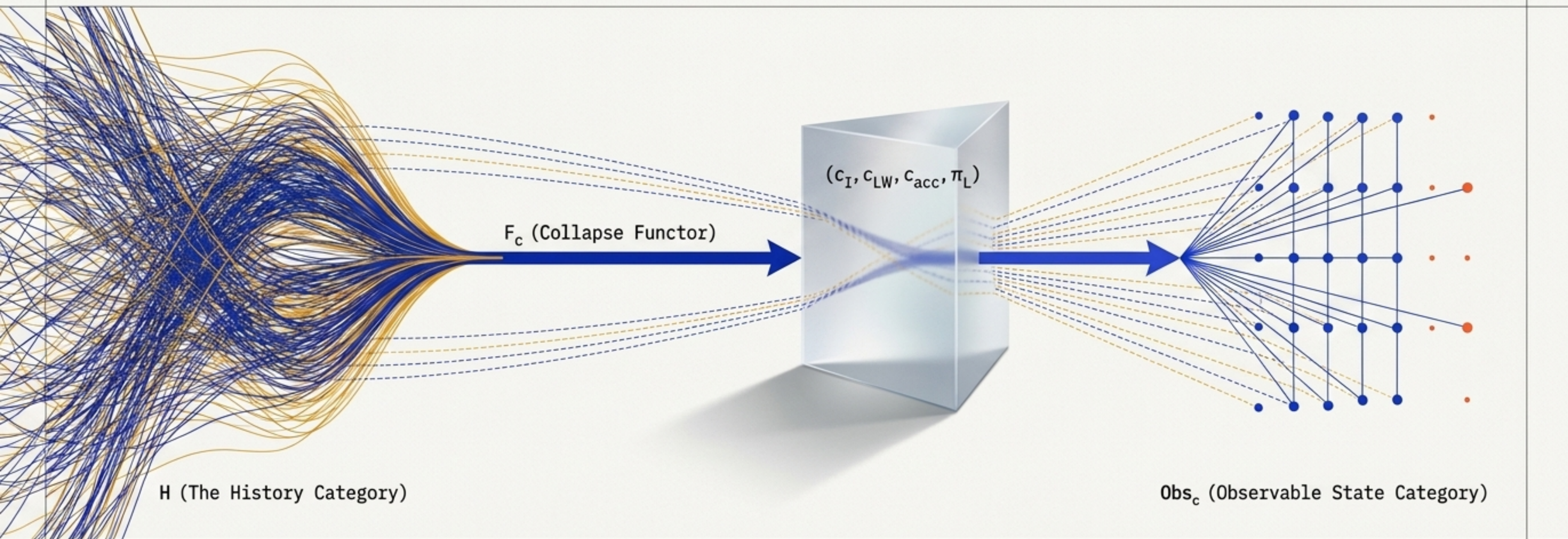
State-centric languages inherit degeneracy as a structural feature.
Spherepop eliminates state erasure by design.

	State (Old World)	History (Spherepop)
Core Unit	Value at a Location	Event Trajectory
Identity	Current Configuration	Unique Factorization of Path
Bug Origins	Race Conditions, Use-after-free	Inadmissible Paths 
Garbage Collection	Structural Unreachability	Admissibility-guided Collapse
Compiler Truth	Output Equivalence	Replay Equivalence
Error Handling	Unwinding / Crashes	Certified Refusal

State-centric languages inherit degeneracy as a structural feature.
Spherepop eliminates state erasure by design.

The Observation Limit: Functors and Projections

The No Direct Observation Theorem: No system with a finite observable space can directly verify its own history without quotienting. Spherepop does not abolish the state illusion—it makes the quotient explicit, parameterized, and history-preserving. We replace implicit erasure with mathematical projection.



The Periodic Table of Artifacts (Note Taxonomy)

Notes are not passive records. They are anti-collapse artifacts designed to preserve access to future continuations. Every artifact maps exactly to a combination of the four fundamental operators.

Note Class	Dominant Operator	Prevents
Capture	Refuse	Experiential collapse (e.g., Photographs, lab notebooks)
Prospective	Bind	Intention-execution gap (e.g., Blueprints, code)
Repair	Refuse + Pop	Procedural interruption (e.g., Troubleshooting guides)
Coordination	Bind	Coordination failure (e.g., Calendars)
Refusal	Refuse	Distinction collapse (e.g., Math proofs, checksums)
Generative	Pop	Inferential poverty (e.g., Maps, languages)
Collapse	Collapse	Navigational failure (e.g., Indexes, taxonomies)

The Ephemerality Inversion

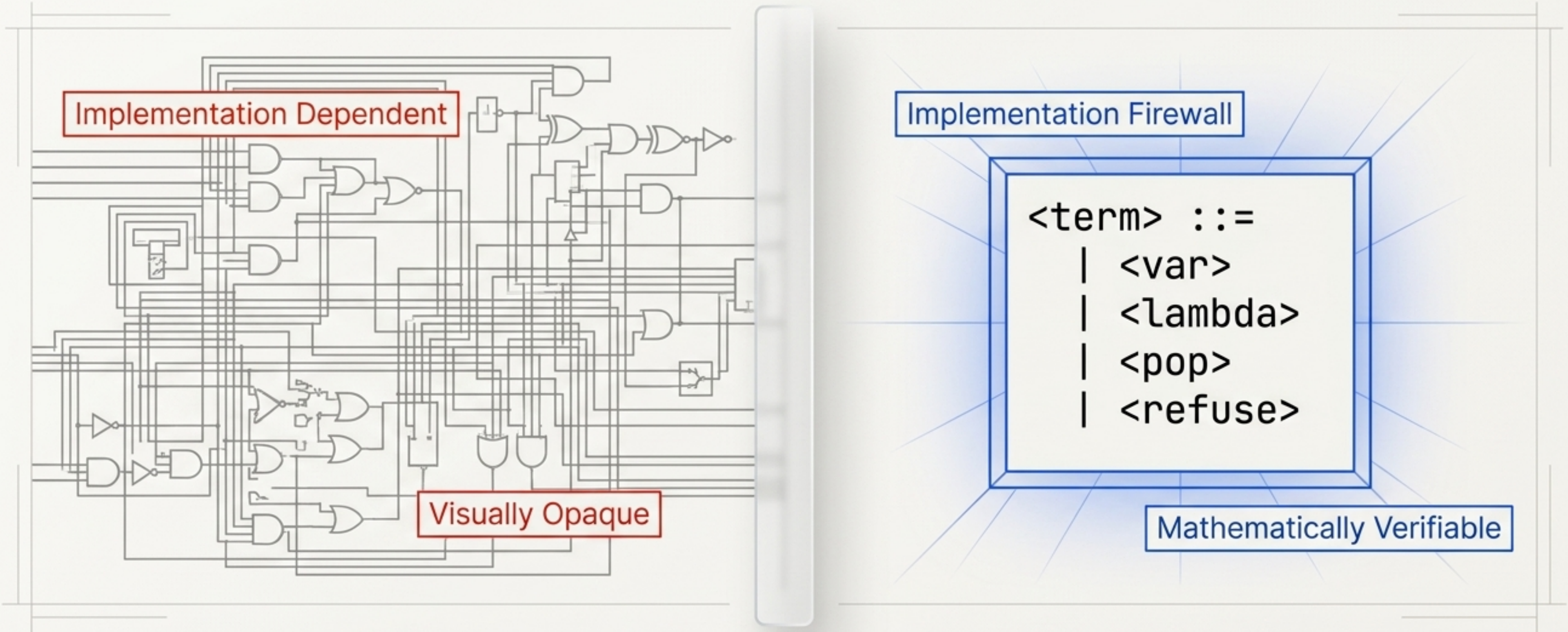
According to the Continuation Completion theorem, a note's value is based on the futures it makes reachable.

Therefore, a destroyed note (if its continuation is executed) represents total success.

Perfect preservation often merely indicates an abandoned trajectory.



Grammar Beats Circuits: The Ultimate Anti-Collapse Artifact



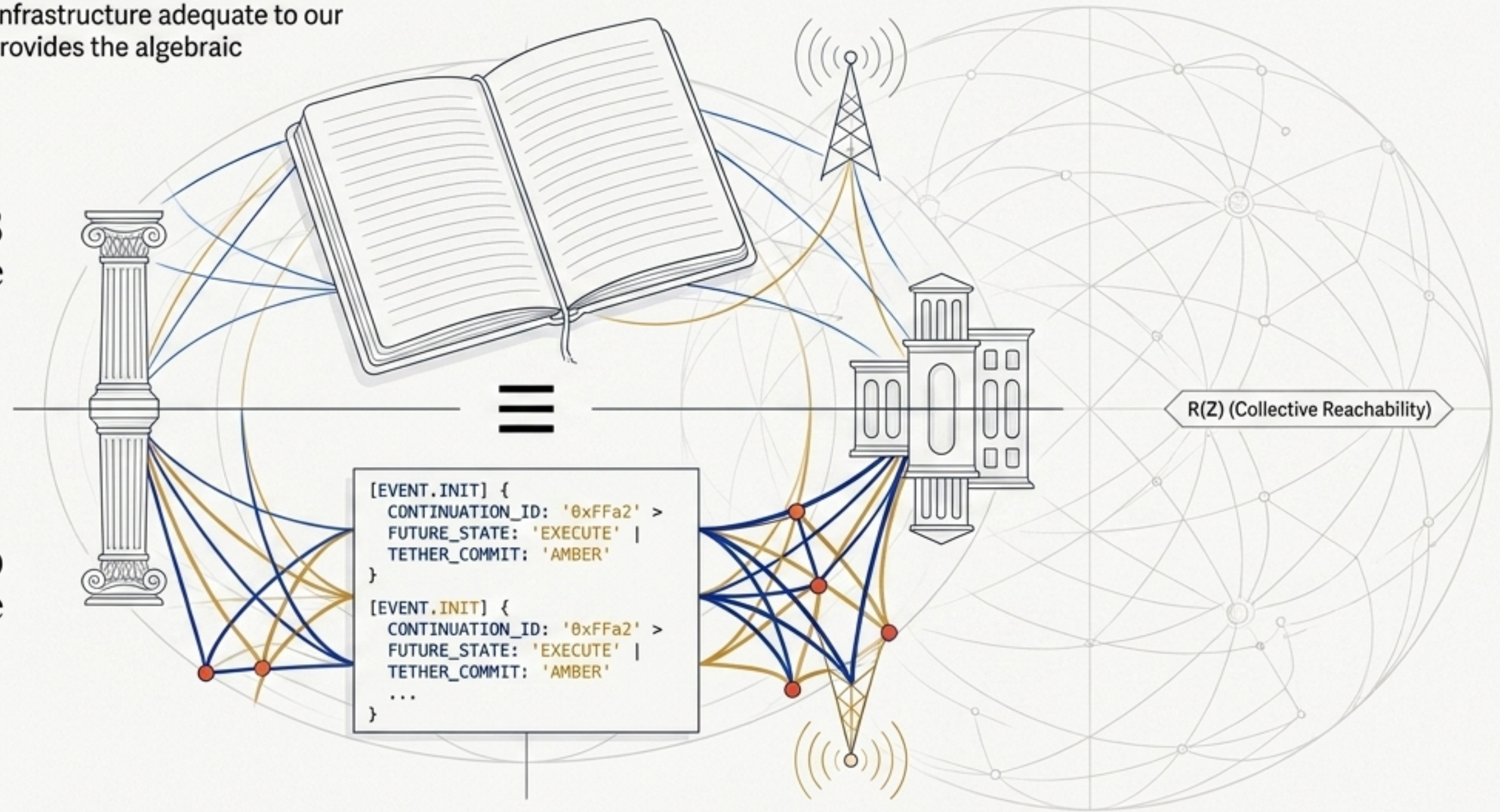
A BNF grammar is a Refusal Note of the highest generativity. It separates shape from semantics, providing an implementation-independent contract. It refuses all invalid strings and generates the full continuation space of all possible valid programs.

Civilization as Note Infrastructure

A Note is an externalized continuation. A Spherepop Program is an internalized note.
Civilization is the attempt to build note infrastructure adequate to our continuation needs; Spherepop finally provides the algebraic substrate to mechanize it.

MEM|8
Cognitive Substrate

Spherepop
Architecture



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